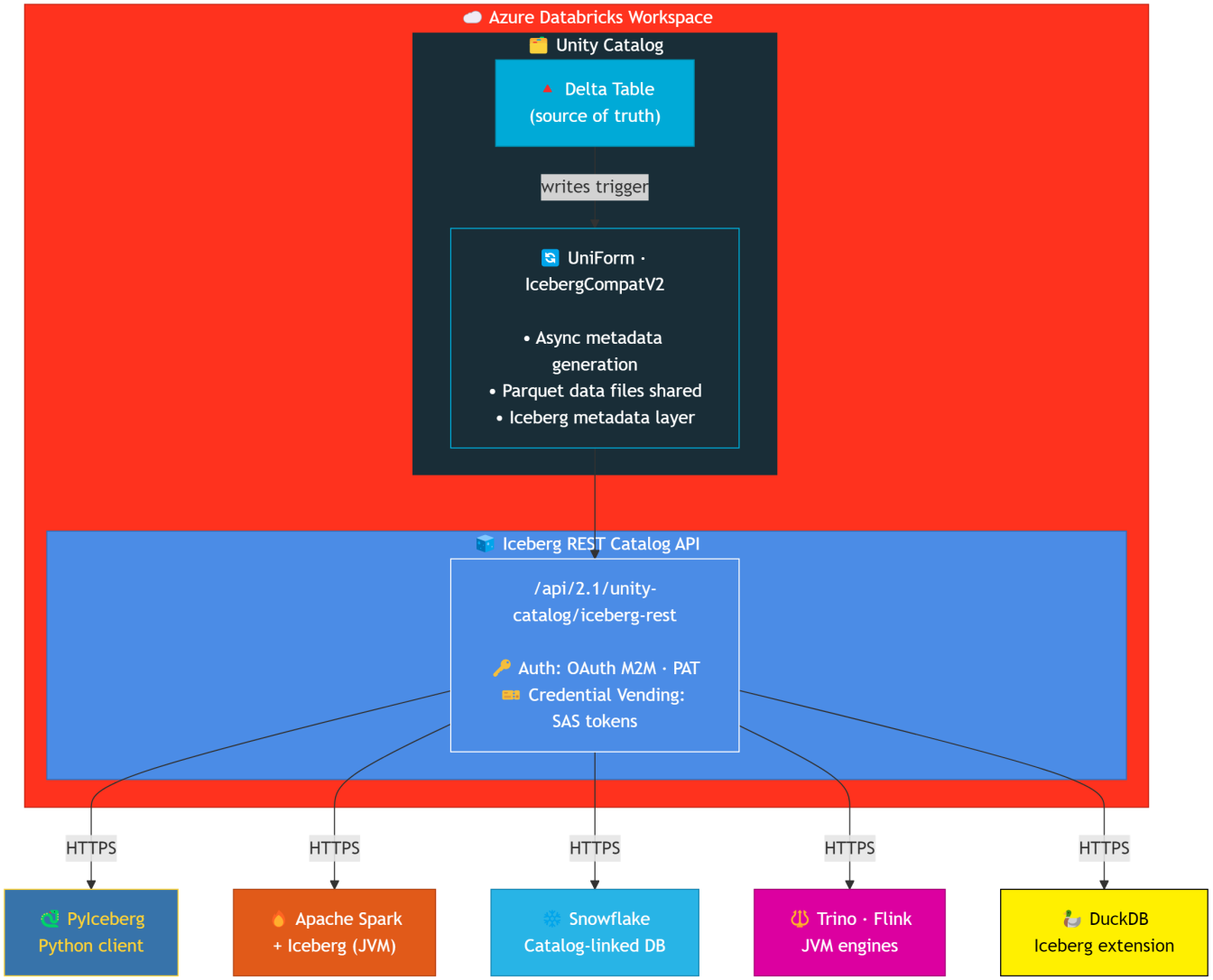
Closure Note — Investigate Iceberg REST Catalog API Feasibility

ADO Ticket: #186438 **Area**: newUM\Data Team **Sprint**: TBD **Priority**: P2 **Sources**: All claims verified against official Microsoft Learn docs (fetched 2026-03-23) AND validated against TEST workspace API (2026-03-24). See References for full list. **Project context**: `clients/oncohealth/knowledge.json` v1.10.0 — operational facts cited as [K].

Architecture Overview

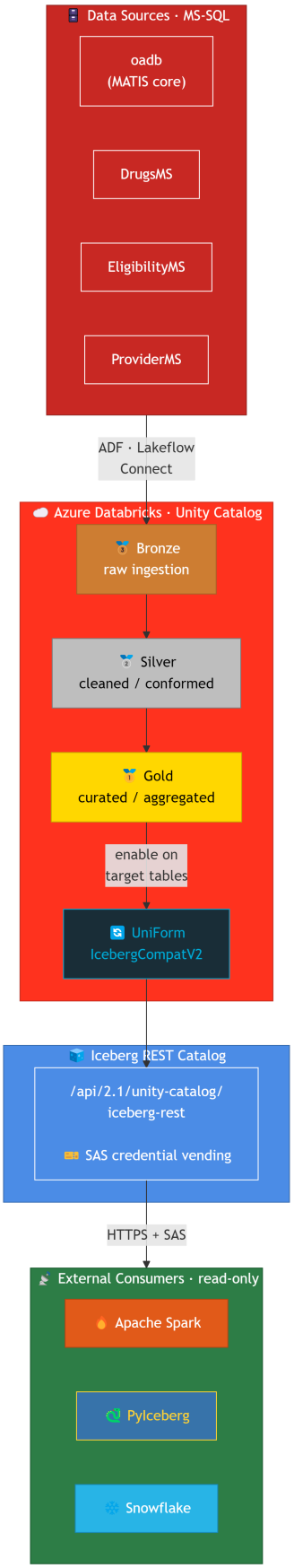


▶ Mermaid source (click to expand)

Data flow: Writes go through Databricks (Delta) → UniForm generates Iceberg metadata async → External clients read via REST Catalog API → Credential vending provides temporary ADLS SAS tokens.

> *Sources: [S1] endpoint + credential vending; [S2] UniForm async metadata generation.*

Applied to newUM



▶ Mermaid source (click to expand)

The above architecture maps to our confirmed environment [K: `tech_stack.data`, `environments.databricks`]:

Layer	newUM Reality	Source
Delta tables	Unity Catalog with Bronze/Silver/Gold medallion	[K: `tech_stack.data`]
Workspace (DEV)	`https://adb-3806388400498653.13.azuredatabricks.net/` — BLOCKED (contains PHI)	[K: `databricks.workspaces[0]`]

Workspace (TEST)	`https://adb-2393860672770324.4.azuredatabricks.net/` — access GRANTED via ticket #0035611 (2026-03-23)	[K: `databricks.workspaces[1]`]
Workspace (UAT/PROD)	URLs unknown	[K: `databricks.workspaces[2-3]`]
PAT token	`visualstudio-carlos` created, expires 2027-03-22	[K: `access_inventory.results["Databricks test"]`]
Auth method	Entra ID button click required (Okta SSO does NOT auto-login to Databricks)	[K: `access_status.access_inventory.note`]
Data Team Lead	Michal Mucha — runs daily standups, Databricks Lakeflow Connect chat	[K: `key_contacts["Michal Mucha"]`]
DevOps contact	`devopsrequest@oncologyanalytics.com`	[K: `onboarding_documents.devops_email`]
Target databases	oadb (main), DrugsMS, EligibilityMS, ProviderMS — all MS-SQL, candidates for lakehouse migration	[K: `environments.databases`]

Cost Estimation

Component	Cost Impact
Iceberg metadata generation	Runs on same compute as Delta writes — marginal increase in driver resource usage
Storage	Iceberg metadata files stored alongside Delta metadata — negligible (~KB per version)
API calls	REST Catalog API calls — included in Databricks pricing, no extra charge
Service principal	No additional licensing cost
Network	If Private Link required — Azure Private Link charges apply
Total incremental cost	**Near-zero** — no new compute or storage required

> ***Note***: *Cost estimation is inferred from architecture described in [S2]. Databricks docs state metadata generation "might increase the driver resource usage" [S2] but provide no cost figures. For high-throughput write workloads, driver memory impact should be validated in TEST.*

Investigation Files

- Full report: `clients/oncohealth/tickets/186438-iceberg-rest-catalog/output.md`

Pros

- **Zero data duplication** — Iceberg reads use same Parquet files as Delta; only metadata is generated ([S2]: "*A single copy of the data files serves multiple formats*")
- **Native Azure Databricks support** — endpoint is built-in, no external infrastructure needed ([S1]: "*Unity Catalog provides an implementation of the Iceberg REST catalog API*")
- **Broad client compatibility** — Pyiceberg, Spark, Snowflake, Trino, Flink supported per [S1]; DuckDB via its own Iceberg extension (not in Databricks docs)
- **Credential vending** — temporary SAS tokens issued automatically, default 1h expiry ([S1]: "*The default expiration time is one hour*")
- **Low operational overhead** — metadata generation is automatic and async ([S2]: "*asynchronously after a Delta Lake write transaction completes*")
- **Standards-based** — uses official Apache Iceberg REST Catalog spec ([S1])
- **Public Preview (DBR 16.4+)** — on GA track, production-viable with preview caveats ([S1]: "*Public Preview in Databricks Runtime 16.4 LTS and above*")

Cons

- **Read-only for Delta+UniForm tables** — external clients cannot write; writes must go through Databricks ([S2]: "*Iceberg client support is read-only. Writes are not supported.*")
- **Metadata lag** — Iceberg metadata generated asynchronously; may lag behind latest Delta version ([S2]: "*Delta table versions do not align with Iceberg versions*")
- **Protocol upgrade partially irreversible** — Iceberg reads can be disabled by unsetting `delta.universalFormat.enabledFormats`, but Delta protocol version upgrades and column mapping **cannot** be undone ([S2]:

"*You can turn off Iceberg reads by unsetting the delta.universalFormat.enabledFormats table property. Upgrades to Delta Lake reader and writer protocol versions cannot be undone.*")

- **Deletion vectors incompatible with Iceberg v2** — tables need `REORG` before enabling; however, **Iceberg v3 supports deletion vectors** ([S2]: "*Apache Iceberg v3 supports deletion vectors*")
- **Public Preview status** — not yet GA; breaking changes possible (low risk given timeline) ([S1])
- **Snowflake+Entra requires public networking** — cannot use Private Link for Entra OAuth ([S1]: "*must use public networking when authenticating with an Entra service principal*")

Risks & Open Questions

#	Risk/Question	Severity	Mitigation
1	**External data access not enabled** — `external_access_enabled: false` on metastore	**HIGH**	Requires UC Admin to enable; escalate via Erik Hjortshoj
2	**No UniForm on any table** — all tables at `minReaderVersion=1`, `minWriterVersion=2`. Protocol upgrade needed on every candidate table.	**HIGH**	Coordinate with Data Team; test on non-critical table first
3	**No `EXTERNAL_USE_SCHEMA` grant** — `NewFire Offshore DBX Users` has SELECT but not EXTERNAL_USE_SCHEMA.	**HIGH**	Request grant from UC Admin
4	**Metadata staleness** — Iceberg metadata may lag Delta writes	MEDIUM	Monitor `converted_delta_version`; use `MSCK REPAIR TABLE` if needed
5	**Network restrictions** — DEV workspace BLOCKED (PHI); TEST accessible via PAT	MEDIUM	Validate firewall/VNet/Private Link config; contact DevOps
6	**Protocol upgrade partially irreversible** — Iceberg reads can be toggled off, but Delta protocol versions and column mapping cannot be undone ([S2])	LOW	Test on non-prod table first; protocol is forward-compatible
7	**Deletion vectors on existing tables** — need REORG for Iceberg v2; Iceberg v3 supports them natively ([S2])	LOW	Schedule during maintenance window; REORG is idempotent
8	**UAT/PROD URLs** still unknown	LOW	Resolve with Michal or DevOps before proceeding

Recommended Next Steps

1. **Coordinate with Michal Mucha** — message sent 2026-03-24, awaiting table selection. Candidates: `drugmaster_test.drug_master.gold_*`, `newum_migration_test.drugs.*`. **WAITING**
2. **Request UC Admin actions** (3 blockers) — message sent to Michal 2026-03-24, awaiting escalation. **WAITING**
   - a. Enable `external_access_enabled` on metastore (`30737b7a-18b6-4e81-9016-03e2c816cc37`)
   - b. Grant `EXTERNAL USE SCHEMA` on target schemas
   - c. Enable UniForm on target table(s) (requires cluster with DBR 14.3+ LTS)
3. **POC on test table** — enable UniForm on selected table, validate via Iceberg REST endpoint
4. **Service principal** — evaluate reusing existing `app-cc28t0 new-data-api` SP; if not, create dedicated SP via DevOps
5. **PyIceberg read test** — after UniForm + external access enabled, validate from outside the workspace
6. **UAT/PROD URLs** — resolve with DevOps

References

All claims in this document were verified against official Microsoft Learn documentation, fetched 2026-03-23:

ID	Title	URL	Last Updated
[S1]	Access Azure Databricks tables from Apache Iceberg clients	https://learn.microsoft.com/en-us/azure/databricks/external-access/iceberg	2026-03-19
[S2]	Read Delta tables with Iceberg clients (UniForm)	https://learn.microsoft.com/en-us/azure/databricks/delta/uniform	2026-03-06
[S3]	Enable external data access on the metastore	https://learn.microsoft.com/en-us/azure/databricks/external-access/admin#external-data-access	—

[S4]	Databricks service principals / Auth	https://learn.microsoft.com/en-us/azure/databricks/dev-tools/auth/	—
[S5]	Pylceberg REST catalog configuration	https://py.iceberg.apache.org/configuration/#rest-catalog	—
[S6]	Apache Iceberg REST API spec	https://github.com/apache/iceberg/blob/master/open-api/rest-catalog-open-api.yaml	—
[K]	Project knowledge base	`clients/oncohealth/knowledge.json` v1.10.0	2026-03-24
[DB]	Databricks TEST workspace API capture	`clients/oncohealth/output/databricks/` (7 files, 987 KB)	2026-03-24